EV Battery Course — Module 7: Battery Management System (BMS)

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The BMS is the battery's brain and bodyguard: it measures every cell, estimates what the battery can do right now, keeps cells balanced, and disconnects the pack before anything unsafe happens. This module covers its hardware, its estimation algorithms and its safety role, and ends with a small simulation you can run yourself.

1. Learning objectives

- Explain why a Li-ion pack cannot operate safely without a BMS.
- Describe what the BMS measures and how (voltage, current, temperature, isolation).
- Define SOC, SOH, SOP and SOE, and explain how each is estimated.
- Explain coulomb counting, OCV lookup, equivalent-circuit models and Kalman filtering.
- Compare passive and active balancing.
- Describe protection functions, contactor control, diagnostics, CAN integration, functional safety and cybersecurity.
- Explain where SOC errors come from and how engineers validate estimators.

2. Fundamentals: why a BMS is needed

Li-ion cells have a narrow safe window. Too high a voltage (overcharge) decomposes electrolyte and can trigger thermal runaway. Too low (overdischarge) dissolves copper from the anode. Too hot accelerates ageing and risks runaway; too cold during charging plates lithium. Cells also drift apart over time. A BMS keeps every cell inside its window, all the time, and tells the vehicle how much energy and power are available.

Typical LFP cell limits (illustrative)

Parameter	Lower limit	Upper limit
Voltage	2.5 V	3.65 V
Discharge temperature	−20 °C	55–60 °C
Charge temperature	0 °C (reduced rates near 0–10 °C)	45–50 °C
Current	Charge and discharge limits from a temperature/SOC table	—

3. BMS architecture

```
flowchart TB
  subgraph PACK["Battery pack"]
    CMU1["Cell monitoring unit 1<br/>voltages + temps of cells 1–16"]
    CMU2["Cell monitoring unit 2<br/>cells 17–32"]
    CMUn["... CMU n"]
    SENSE["Current sensor, pack voltage, isolation monitor, HVIL"]
  end
  CMU1 -- "isolated daisy-chain or wireless" --> BMU["Battery management unit (master)<br/>state estimation, protection, balancing, contactor control"]
  CMU2 --> BMU
  CMUn --> BMU
  SENSE --> BMU
  BMU --> CONT["Contactors and pre-charge"]
  BMU -- "CAN" --> VCU["Vehicle control unit, charger, thermal system, dashboard"]
```

- **Centralised:** one board with wires to every cell. Cheap for small packs (two-wheelers); heavy harness for large packs.
- **Distributed (master–slave):** cell monitoring units (CMUs) on each module, linked by an isolated communication chain to a master. Most EV cars.
- **Wireless BMS:** CMUs communicate by radio, removing harness and connectors (GM Ultium uses a wireless BMS).

4. What the BMS measures

Measurement	How	Typical accuracy needed
Cell voltage	Dedicated monitoring ICs (e.g. from ADI, TI, NXP, Infineon), 12–18 cells per chip	±1–2 mV (critical for LFP's flat curve)
Pack current	Shunt resistor (precise, needs temperature compensation) or Hall sensor (isolated, offset drift)	Offset < ~0.1–0.5 A; gain error < ~0.5%
Temperature	NTC thermistors on cells, busbars, coolant	±1–2 °C
Pack voltage (both sides of contactors)	Isolated measurement	For pre-charge and contactor diagnostics
Isolation resistance	Injected signal between HV and chassis	Detect faults above ~100 Ω/V threshold

Not every cell has a thermistor; engineers place them at predicted hot and cold spots.

5. The states the BMS estimates

State	Meaning	Unit	Used for
SOC (state of charge)	Remaining charge as % of current full capacity	%	Fuel gauge, charge control
SOE (state of energy)	Remaining energy	kWh	Range estimate (SOC × voltage is not linear)
SOH (state of health)	Current capacity (and resistance) vs new	%	Warranty, ageing, second life
SOP (state of power)	Max power available for the next ~2–30 s without violating limits	kW	Motor and charger power limits

Coulomb counting

```
SOC(t) = SOC(t_0) - \frac{1}{3600\,C_{nom}} \int_{t_0}^{t} \eta \, I(\tau)\, d\tau
```

I in A (positive = discharge), C_nom in Ah, η = coulombic efficiency (~1 for Li-ion). Simple and responsive, but integrates every sensor error forever, and depends on knowing the starting SOC and true capacity.

OCV lookup

After rest (typically 30 min to several hours), terminal voltage ≈ OCV, and OCV maps to SOC through a table measured per chemistry and temperature. Accurate for NMC; weak for LFP across 20–90% SOC because the curve is flat (a 1 mV error can mean several % SOC).

Equivalent-circuit model (ECM)

```
flowchart LR
  OCV["OCV(SOC, T)"] --> R0["R0: ohmic resistance"] --> RC1["R1 || C1: charge
transfer (seconds)"] --> RC2["R2 || C2: diffusion (minutes)"] -->
TERM["Terminal voltage"]
```

```
V_t = OCV(SOC,T) - I R_0 - V_1 - V_2 \qquad \frac{dV_k}{dt} = -\frac{V_k}{R_k
C_k} + \frac{I}{C_k}
```

Parameters are fitted from pulse tests across SOC and temperature (Module 11) and stored as lookup tables.

Kalman filtering

A Kalman filter combines two imperfect information sources: a **prediction** (coulomb counting through the model) and a **measurement** (actual terminal voltage). It compares measured voltage with the model's predicted voltage; the difference corrects the SOC estimate, weighted by how much it trusts each source. Because the battery is non-linear, BMS use the **extended Kalman filter (EKF)** or **unscented Kalman filter (UKF)**. Benefits: self-correcting, no need for long rests, estimates uncertainty. Costs: needs a good model and tuning.

SOH and SOP

- **Capacity SOH:** compare Ah counted between two well-known SOC points against nominal capacity; or fit capacity inside the Kalman filter.
- **Resistance SOH:** ΔV ÷ ΔI at current steps (e.g. hard acceleration).
- **SOP:** solve the ECM for the maximum current that keeps the lowest cell above V_min (discharge) or highest cell below V_max (charge) for the chosen time window, then cap by current and temperature limits.

6. Cell balancing

Cells drift apart because of differences in self-discharge, capacity and temperature. Without balancing, the highest cell reaches V_max first and stops charging while others are not full; the lowest stops discharge early. Usable capacity shrinks.

Feature	Passive balancing	Active balancing
Method	Bleeds excess charge from high cells through a resistor	Moves charge from high cells to low cells (capacitor, inductor or transformer)
Current	Typically 50–200 mA	Up to several A
Efficiency	Energy is wasted as heat	Most energy is kept (about 80–95% transfer efficiency)
Cost and complexity	Low	High
Usage	Almost all EVs today	Some premium, second-life and storage systems

Balancing is usually done near top of charge (where cell differences are visible on the voltage curve), or using SOC differences calculated by the BMS.

7. Protection, contactors and diagnostics

Fault	Typical response
Cell overvoltage	Stop charging / reduce regen; open contactors at a hard limit
Cell undervoltage	Reduce power, then stop discharge
Overcurrent	Limit via SOP; fuse/pyrofuse for short circuit
Over-temperature	Derate power; request cooling; open contactors at hard limit
Under-temperature charging	Block or limit charge; request heating
Isolation fault	Warn, then prevent next start (depending on severity)
HVIL open	Open contactors

Fault	Typical response
Sensor or communication loss	Fall back to safe limits, warn driver
Contactor welded	Detect by voltage across contactor; store fault; warn

The BMS logs faults with time, SOC, temperatures and cell voltages ("freeze frame") for diagnostics, warranty analysis and field recalls.

8. Vehicle integration, hardware, firmware and calibration

- **CAN (Controller Area Network):** the BMS broadcasts SOC, SOP limits, cell min/max, temperatures and faults; receives commands (wake, drive, charge, contactor requests) from the VCU and charger. Newer vehicles also use CAN FD or automotive Ethernet.
- **Charger communication:** DC fast charging uses a protocol between vehicle and charger (CCS2 uses PLC per ISO 15118 / DIN 70121; Bharat DC-001 and GB/T use CAN). The BMS requests current and voltage.
- **Firmware layers:** low-level drivers and safety monitoring, state estimation, control (balancing, thermal requests), communications, diagnostics, bootloader for over-the-air updates.
- **Calibration:** OCV tables, ECM parameters, current/temperature limit maps, thermal requests and derating curves, tuned per cell and pack.

9. Functional safety and cybersecurity

- **ISO 26262** governs functional safety of automotive electronics. Battery hazards like overcharge leading to fire are typically rated ASIL C or D, requiring redundancy (e.g. independent secondary voltage monitoring), diagnostic coverage and a safe state (contactors open).
- **ISO/SAE 21434 and UNECE R155/R156** govern cybersecurity and software updates. The BMS must authenticate firmware updates and resist malicious CAN messages, because a hacked BMS could disable protections.

10. Sample cell-monitoring table (LFP, 16-cell module, during charge)

Cell	Voltage (V)	Temp (°C)	Status
1	3.412	31.2	OK
2	3.415	31.8	OK
3	3.409	32.0	OK
4	3.468	32.5	Highest; balancing ON
5	3.411	32.4	OK
6	3.402	33.9	Lowest voltage; warmest; watch
7–16	3.408–3.417	31.0–33.1	OK
Module	min 3.402 / max 3.468 / spread 66 mV	max 33.9	Charge current limited by cell 4

Reading it: a 66 mV spread near top of charge in LFP signals imbalance (here, cell 4 is fuller). Cell 6 being both lowest and warmest is worth tracking: higher resistance or self-discharge may be developing.

11. Why SOC estimates go wrong, and how engineers validate them

Error sources

- Current sensor offset and gain error (integrates over time).
- Unknown initial SOC after long parking without enough rest.
- Capacity used in the calculation is outdated (pack has aged).
- OCV hysteresis (LFP OCV differs after charge vs discharge).
- Temperature effects and inaccurate model parameters.
- Cell imbalance: pack SOC is not the SOC of every cell.

Validation

1. Lab: run the cell or pack on a high-precision cycler through drive cycles (WLTC, MIDC, aggressive profiles) across temperatures. The cycler's precise Ah counting plus full charge/discharge endpoints give "true" SOC.

2. Hardware-in-the-loop (HIL): run the real BMS against a simulated pack, including faults and sensor errors.
3. Vehicle: compare predicted vs actual remaining energy at end of trips; check range at low SOC.
4. Typical target: SOC error within about ±3–5% across conditions (OEM-specific), tighter at low SOC where it matters most.

12. Practical project: coulomb counting with simulated data

This Python script simulates 4 hours of driving at 10–20 A on a 160 Ah pack, with a current sensor that reads 0.5 A too high plus random noise. It compares true SOC with the BMS's coulomb-counted SOC.

```
import random
random.seed(1)                # repeatable random numbers
C_ah = 160.0                  # pack capacity in Ah
dt = 1.0                      # time step: 1 second
soc_true = 0.90               # real starting SOC
soc_cc = 0.90                 # BMS estimate starts correct
bias = 0.5                    # current sensor offset, A
for t in range(4 * 3600):     # 4 hours, one loop per second
    I = 10 + 10 * random.random()                  # true discharge current
10–20 A
    I_meas = I + bias + random.gauss(0, 1.0)        # what the sensor reports
    soc_true -= I * dt / 3600 / C_ah                # true SOC falls by Ah
used / capacity
    soc_cc   -= I_meas * dt / 3600 / C_ah           # BMS integrates the
measured current
print(f"true {soc_true*100:.2f}%  counted {soc_cc*100:.2f}%  error {(soc_cc-
soc_true)*100:.2f} pts")
```

Result when run: true SOC 52.46%, counted 51.21%, error **−1.24 percentage points** after 4 hours.

What it shows: random noise averages out, but a constant offset does not. 0.5 A ÷ 160 Ah = 0.31% SOC error per hour, so 4 h → 1.25 points. Left alone for days, the error keeps growing, which is why real BMS fuse coulomb counting with voltage-based correction (OCV after rest or a Kalman filter).

Try next: add a simple correction (reset soc_cc to the true value whenever the pack "rests"), or reduce the capacity value in the BMS to 150 Ah to mimic an aged pack the BMS doesn't know about.

13. Real-world EV examples

- GM Ultium: wireless BMS reduces wiring by an estimated ~90% in the pack (GM's claim).
- Tesla: in-house BMS with frequent over-the-air updates, including charging-curve and range-estimate changes.
- Indian two-wheelers: many makers moved BMS design in-house after early fire incidents to control protection logic and data logging.
- LFP vehicles (BYD, Tata): owner's manuals recommend periodic full charges so the BMS can recalibrate SOC.

14. Manufacturing applications

- **EOL calibration:** current sensor offset and gain calibrated per pack; voltage channels checked against a reference meter.
- **Flash and configuration:** each pack's BMS is programmed with firmware and the correct cell/pack configuration; traceability links serial numbers.
- **Initial balancing:** packs may leave the line with cells balanced at a known SOC.
- **Measurement system analysis:** gauge R&R on the EOL voltage and current test equipment (Module 11).

15. Common mistakes and misconceptions

- **"SOC = voltage."** Only at rest, and poorly for LFP.
- **"The BMS balances during driving."** Mostly at top of charge or when parked, depending on design.
- **"Passive balancing fixes weak cells."** It only equalises charge; it cannot restore lost capacity.
- **"Remaining range is calculated from SOC only."** It also uses recent consumption, temperature and SOE.
- **"BMS software is fixed after production."** OTA updates change limits and estimation in the field.

16. Practical case study: range drops suddenly at low SOC

Problem. Drivers report the range jumping from 40 km to 10 km in a few minutes in winter.

Root cause. The BMS used SOC rather than SOP/SOE for range at low temperature. Cold cells had higher resistance, so the lowest cell hit V_min under load while SOC still showed 15%; the BMS corrected SOC abruptly.

Fix. Use temperature-dependent ECM parameters, estimate SOE with expected load, add a temperature-aware reserve, and smooth displayed SOC.

Lesson. Estimation quality is a customer-trust issue, not just a technical one.

17. Revision notes

- BMS = measure, estimate, protect, balance, communicate.
- SOC (charge), SOE (energy), SOH (ageing), SOP (power now).
- Coulomb counting drifts; OCV needs rest; Kalman filters fuse both through a model.
- Passive balancing wastes; active balancing moves charge.
- ISO 26262 for safety, ISO/SAE 21434 and UNECE R155 for cybersecurity.

18. Glossary

Term	Meaning
CMU / BMU	Cell monitoring unit / battery management unit (master)
ECM	Equivalent-circuit model
EKF / UKF	Extended / unscented Kalman filter
ASIL	Automotive Safety Integrity Level (A lowest, D highest)
OTA	Over-the-air software update
HIL	Hardware-in-the-loop testing
Hysteresis	OCV depending on whether the cell was last charged or discharged

19. Knowledge check

Conceptual

1. Why is coulomb counting alone not enough over weeks of use?
2. Why is cell voltage accuracy more critical for LFP than NMC?

3. What is the difference between SOC and SOE, and why does range use SOE?
4. When does passive balancing work best, and why?
5. Why would a BMS overcharge hazard be rated ASIL C or D?

Numerical

1. A 100 Ah pack's current sensor has a +0.3 A offset. What SOC error builds up in 10 hours of operation?
2. A balancing resistor bleeds 100 mA. How long to remove a 1 Ah imbalance? How much energy is wasted at 3.3 V?
3. ECM: OCV 3.30 V, R0 = 0.5 mΩ, V1 = 15 mV, V2 = 10 mV, I = 200 A discharge. Find terminal voltage.

Application

1. A scooter maker sees SOC jump from 20% to 5% during hill climbs. Propose three BMS-level fixes.
2. As a costing engineer, compare wired distributed BMS with wireless BMS for a 120-cell pack.

20. Answer key

Conceptual

1. It integrates current-sensor offset and gain errors indefinitely, has no way to correct the starting SOC, and uses a capacity that becomes outdated as the pack ages.
2. LFP's OCV changes only a few mV across most of its SOC range, so a few mV of measurement error equals several % SOC; NMC changes far more per % SOC.
3. SOC is charge (Ah); SOE is energy (Wh). Energy per Ah falls as voltage drops at low SOC and with load and temperature, so SOE predicts range better.
4. Near top of charge, where cell voltages diverge noticeably and imbalance is visible; LFP's flat mid-range hides differences.
5. Overcharge can cause thermal runaway and fire, with severe injury possible (high severity), during common operation (high exposure), with limited driver ability to control it.

Numerical

1. 0.3 × 10 = 3 Ah → 3 ÷ 100 = **3% SOC error**.
2. 1 Ah ÷ 0.1 A = **10 hours**. Energy = 1 Ah × 3.3 V = **3.3 Wh** wasted as heat.
3. $V_t = 3.30 - 200 \times 0.0005 - 0.015 - 0.010 = 3.30 - 0.100 - 0.025 =$ **3.175 V**.

Application

1. Use SOP and model-based estimation (EKF) so voltage sag under load isn't misread as low SOC; use temperature- and age-dependent resistance; smooth displayed SOC and add a reserve; balance cells better if the lowest cell is triggering it; update capacity for ageing.
2. Wireless removes the communication harness, connectors and their assembly labour and failure modes, saves mass and eases second-life reuse; it adds radio modules, certification, security work and possibly licensing cost. For 120 cells (about 8 CMUs), compare harness + connector + labour + warranty cost per pack against radio BOM; wireless gains more as pack variants multiply.

21. Further learning

- Gregory Plett, *Battery Management Systems, Vol. I: Battery Modeling* and *Vol. II: Equivalent-Circuit Methods* (Artech House); his free University of Colorado course on Coursera.
- Davide Andrea, *Battery Management Systems for Large Lithium-Ion Battery Packs*.
- ISO 26262 (functional safety); ISO/SAE 21434 (cybersecurity).
- Texas Instruments and Analog Devices BMS application notes (free).

Next: Module 8, battery thermal management.